

Homework #3 - MATH 257

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Problem 1

We have four I.I.D. multivariate normal random variables $\mathbf{X}_1, \mathbf{X}_2, \mathbf{X}_3, \mathbf{X}_4 \sim N_p(\mu, \Sigma)$

Part (a)

For $\mathbf{V}_1 = \frac{1}{4}\mathbf{X}_1 - \frac{1}{4}\mathbf{X}_2 + \frac{1}{4}\mathbf{X}_3 - \frac{1}{4}\mathbf{X}_4$, since $\mathbf{V}_1$ is a linear combination of I.I.D. normally distributed random variables, then $\mathbf{V}_1$ is also normally distributed.

Using Result 4.8 from Johnson and Wichern, the distribution of this linear combination will leverage the following formula:

$$\mathbf{V} = \sum_{i=1}^n c_i \mathbf{X}_i \sim N_p\left(\sum_{i=1}^n c_i \mu, \left(\sum_{i=1}^n c_i^2\right) \Sigma\right)$$

We evaluate the mean and variance respectively:

$$\sum_{i=1}^n c_i \mu = \frac{1}{4}\mu - \frac{1}{4}\mu + \frac{1}{4}\mu - \frac{1}{4}\mu = 0$$

$$\left(\sum_{i=1}^n c_i^2\right) \Sigma = \frac{1}{16}\Sigma + \frac{1}{16}\Sigma + \frac{1}{16}\Sigma + \frac{1}{16}\Sigma = \frac{1}{4}\Sigma$$

Therefore, $\mathbf{V}_1 \sim N_p\left(0, \frac{1}{4}\Sigma\right)$.

By a similar calculation, $\mathbf{V}_2 \sim N_p\left(0, \frac{1}{4}\Sigma\right)$.

We notice that $\mathbf{V}_1$ and $\mathbf{V}_2$ have the exact same distribution even though they have different linear combinations. This is reasonable, however, as $\mathbf{V}_1$ can be transformed into $\mathbf{V}_2$ if one swaps the order of the $\mathbf{X}_1$ and $\mathbf{X}_2$ terms, and relabels $\mathbf{X}_1$ as $\mathbf{X}_2$, and vice versa. Since every $\mathbf{X}_i$ follows the exact same distribution, then distributional form of $\mathbf{V}_1$ and $\mathbf{V}_2$.

Part (b)

For part (b), we reference Result 4.5(c) from Johnson and Wichern:

Theorem 0.1 (Joint Multivariate Distribution). *If $\mathbf{X}_1$ and $\mathbf{X}_2$ are independent and are distributed as $N_{q_1}(\boldsymbol{\mu}_1, \boldsymbol{\Sigma}_{11})$ and $N_{q_2}(\boldsymbol{\mu}_2, \boldsymbol{\Sigma}_{22})$, then*

$$\begin{bmatrix} \mathbf{X}_1 \\ \mathbf{X}_2 \end{bmatrix} \sim N_{q_1+q_2} \left(\begin{bmatrix} \boldsymbol{\mu}_1 \\ \boldsymbol{\mu}_2 \end{bmatrix}, \begin{bmatrix} \boldsymbol{\Sigma}_{1,1} & \mathbf{0} \\ \mathbf{0} & \boldsymbol{\Sigma}_{2,2} \end{bmatrix} \right)$$

Before using the result straightaway, we must first check that $\mathbf{V}_1$ and $\mathbf{V}_2$ are indeed independent. From part (a), since we've shown that they are normally distributed, if we can that the covariance of $\mathbf{V}_1$ and $\mathbf{V}_2$ is zero, then they are mutually independent.

Using the bilinearity of covariance:

$$\text{Cov}(\mathbf{V}_1, \mathbf{V}_2) = \sum_{i=1}^4 (v_{1,i} \cdot v_{2,i} \cdot \boldsymbol{\Sigma})$$

where $v_{1,i}$ and $v_{2,i}$ are the coefficients of $\mathbf{V}_1$ and $\mathbf{V}_2$, respectively.

We will now check the entrywise product of the coefficients:

$$v_{1,i} \cdot v_{2,i} = \frac{1}{4} \cdot \frac{1}{4} - \frac{1}{4} \cdot \frac{1}{4} - \frac{1}{4} \cdot \frac{1}{4} + \frac{1}{4} \cdot \frac{1}{4} = \frac{1}{16} - \frac{1}{16} - \frac{1}{16} + \frac{1}{16} = 0$$

Since the entrywise product of the coefficients is equal to zero, the covariance between $\mathbf{V}_1$ and $\mathbf{V}_2$ is also zero. As such, $\mathbf{V}_1$ and $\mathbf{V}_2$ are mutually independent.

Therefore, the join density of $\mathbf{V}_1$ and $\mathbf{V}_2$ is:

$$\begin{bmatrix} \mathbf{V}_1 \\ \mathbf{V}_2 \end{bmatrix} \sim N_{2p} \left(\begin{bmatrix} \mathbf{0} \\ \mathbf{0} \end{bmatrix}, \begin{bmatrix} \frac{1}{4}\boldsymbol{\Sigma} & \mathbf{0} \\ \mathbf{0} & \frac{1}{4}\boldsymbol{\Sigma} \end{bmatrix} \right)$$

Problem 2

Part (a)

$\bar{\mathbf{X}}$ is the sample average of all sixty four-variate random variables $\mathbf{X}_i$. Meaning that it can be written as a linear combination:

$$\bar{\mathbf{X}} = \frac{1}{n} \sum_{i=1}^n \mathbf{X}_i = \frac{\mathbf{X}_1 + \dots + \mathbf{X}_{60}}{60}$$

Therefore, using Result 4.8, and letting $c_i = \frac{1}{60}$:

$$\begin{aligned} \bar{\mathbf{X}} &\sim N_4 \left(\sum_{i=1}^{60} \frac{1}{60} \mu, \left(\sum_{i=1}^{60} \frac{1}{60} \right)^2 \Sigma \right) \\ &\sim N_4 \left(\mu, \frac{1}{60} \Sigma \right) \end{aligned}$$

Part (b)

Shifting $\mathbf{X}_1$ over by μ results in $(\mathbf{X}_1 - \mu) \sim N_4(0, \Sigma)$. $(\mathbf{X}_1 - \mu)^\top \Sigma^{-1} (\mathbf{X}_1 - \mu)$ expresses a sum of four squared normal variables. Therefore, $(\mathbf{X}_1 - \mu)^\top \Sigma^{-1} (\mathbf{X}_1 - \mu) \sim \chi_4^2$

Part (c)

From part (b), $(\mathbf{X} - \mu)^\top \Sigma^{-1} (\mathbf{X} - \mu) \sim \chi_p^2$ if $\mathbf{X}_i$ are IID standard normal.

Since $\bar{\mathbf{X}} \sim N_p \left(\mu, \frac{\Sigma}{60} \right)$, then $(\bar{\mathbf{X}} - \mu) \sim N_p \left(0, \frac{\Sigma}{60} \right)$.

Substituting the sample covariance matrix Σ/n in for the expression in part (b), then:

$$(\bar{\mathbf{X}} - \mu)^\top \left(\frac{\Sigma}{n} \right)^{-1} (\bar{\mathbf{X}} - \mu) = n(\bar{\mathbf{X}} - \mu)^\top \Sigma^{-1} (\bar{\mathbf{X}} - \mu) \sim \chi_4^2$$

Part (d)

The approximate distribution of $n(\bar{\mathbf{X}} - \mu)^\top \mathbf{S}^{-1} (\bar{\mathbf{X}} - \mu)$ is χ_4^2

Problem 3

Part (a)

The formula for the Hotelling's T^2 statistic is the following:

$$T^2 = (\bar{\mathbf{X}} - \boldsymbol{\mu}_0)^\top \left(\frac{1}{n} \mathbf{S} \right)^{-1} (\bar{\mathbf{X}} - \boldsymbol{\mu}_0)$$

Where:

$$\bar{\mathbf{X}} = \frac{1}{n} \sum_{i=1}^n \mathbf{X}_i$$
$$\mathbf{S} = \frac{1}{n-1} \sum_{i=1}^n (\mathbf{X}_i - \bar{\mathbf{X}})(\mathbf{X}_i - \bar{\mathbf{X}})^\top$$

For this problem:

$$\bar{\mathbf{X}} = \begin{bmatrix} \frac{2+8+6+8}{4} \\ \frac{12+9+9+10}{4} \end{bmatrix} = \begin{bmatrix} \frac{24}{4} \\ \frac{40}{4} \end{bmatrix} = \begin{bmatrix} 6 \\ 10 \end{bmatrix}$$
$$\mathbf{S} = \frac{1}{4-1} \sum_{i=1}^4 (\mathbf{X}_i - \bar{\mathbf{X}})(\mathbf{X}_i - \bar{\mathbf{X}})^\top$$
$$= \frac{1}{3} \left[\begin{bmatrix} -4 \\ 2 \end{bmatrix} \begin{bmatrix} -4 \\ 2 \end{bmatrix}^\top + \begin{bmatrix} 2 \\ -1 \end{bmatrix} \begin{bmatrix} 2 \\ -1 \end{bmatrix}^\top + \begin{bmatrix} 0 \\ -1 \end{bmatrix} \begin{bmatrix} 0 \\ -1 \end{bmatrix}^\top + \begin{bmatrix} 2 \\ 0 \end{bmatrix} \begin{bmatrix} 2 \\ 0 \end{bmatrix}^\top \right]$$
$$= \frac{1}{3} \left[\begin{bmatrix} 16 & -8 \\ -8 & 4 \end{bmatrix} + \begin{bmatrix} 4 & -2 \\ -2 & 1 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} 4 & 0 \\ 0 & 0 \end{bmatrix} \right]$$
$$= \frac{1}{3} \begin{bmatrix} 24 & -10 \\ -10 & 6 \end{bmatrix}$$
$$= \begin{bmatrix} 8 & -\frac{10}{3} \\ -\frac{10}{3} & 2 \end{bmatrix}$$

With $\bar{\mathbf{X}}$ and $\mathbf{S}$ now known, we can compute T^2 :

$$\begin{aligned}
T^2 &= \left(\begin{bmatrix} 6 \\ 10 \end{bmatrix} - \begin{bmatrix} 7 \\ 11 \end{bmatrix} \right)^\top \cdot \left(\frac{1}{4} \begin{bmatrix} 8 & -\frac{10}{3} \\ -\frac{10}{3} & 2 \end{bmatrix} \right)^{-1} \left(\begin{bmatrix} 6 \\ 10 \end{bmatrix} - \begin{bmatrix} 7 \\ 11 \end{bmatrix} \right) \\
&= [-1 \quad -1] \cdot \begin{bmatrix} 2 & -\frac{10}{12} \\ -\frac{10}{12} & \frac{1}{2} \end{bmatrix}^{-1} \cdot \begin{bmatrix} -1 \\ -1 \end{bmatrix} \\
&= [-1 \quad -1] \cdot \begin{bmatrix} \frac{18}{11} & \frac{30}{11} \\ \frac{30}{11} & \frac{72}{11} \end{bmatrix} \cdot \begin{bmatrix} -1 \\ -1 \end{bmatrix} \\
&= \boxed{\frac{150}{11}}
\end{aligned}$$

Part (b)

It is known that T^2 follows a scaled F -distribution: $\frac{(n-1)p}{(n-p)} F_{p, n-p}$

For our **particular** T^2 , it follows $T^2 \sim \frac{(4-1) \cdot 2}{4-2} F_{2, 4-2} = 3F_{2, 2}$.

Part (c)

Given the $\alpha = 0.05$ significance level, we test whether or not

$$T^2 \geq 3F_{2, 2}(\alpha = 0.05)$$

If $T^2 \geq 3F_{2, 2}(\alpha = 0.05)$, then we can reject the null hypothesis that the true $\boldsymbol{\mu}$ is equal to $\boldsymbol{\mu}_0 = \begin{bmatrix} 7 \\ 11 \end{bmatrix}$. If $T^2 < 3F_{2, 2}(\alpha = 0.05)$, then we fail to reject the null hypothesis.

Computing $3F_{2, 2}(0.05)$, we get

$$3F_{2, 2}(0.05) = 57$$

Since $T^2 = \frac{150}{11}$, a quantity that is **less than** 57, we fail to reject the null hypothesis.

Problem 4

Part (a)

The James-Stein estimator is defined as the following:

$$\hat{\mu}^{JS}(\mathbf{X}) = \left(1 - \frac{p-2}{\|\mathbf{X}\|^2}\right) \mathbf{X}$$

For a single dataset, we can get a lower MSE loss with the $\mathbf{X}$ as the estimator for θ as demonstrated in the following code:

```
set.seed(12345)
Theta <- c(1, 10, 100)
X <- rnorm(n = 3, mean = Theta, sd = 1)
james_stein <- (1-(1/sum(X^2)))*X

mse_x <- mean((X - Theta) ^ 2)
mse_js <- mean((james_stein - Theta)^2)
paste("Mean-Squared Error of X and Theta:", mse_x)
```

```
[1] "Mean-Squared Error of X and Theta: 0.286044413708904"
```

```
paste("Mean-Squared Error of James-Stein and Theta:", mse_js)
```

```
[1] "Mean-Squared Error of James-Stein and Theta: 0.286235393185314"
```

We see in this case that, in this instance, the MSE loss of the James-Stein estimator is larger than the MSE loss of $\mathbf{X}$.

Part (b)

When we repeatedly sample, however, then calculate the loss function, and then take the average of all those samples (to find the risk function), the James-Stein estimator has a lower risk:

```

set.seed(12345)
Theta <- c(1, 10, 100)

mse_x_list <- numeric(10000)
mse_js_list <- numeric(10000)

for (i in 1:10000)
{
  X <- rnorm(n = 3, mean = Theta, sd = 1)
  james_stein <- (1-(1/sum(X^2)))*X

  mse_x_list[i] <- sum((X - Theta) ^ 2)
  mse_js_list[i] <- sum((james_stein - Theta)^2)
}

risk_x <- mean(mse_x_list)
risk_js <- mean(mse_js_list)

paste("Risk of X and Theta:", risk_x)

```

```
[1] "Risk of X and Theta: 2.97667504104805"
```

```
paste("Risk of James-Stein and Theta:", risk_js)
```

```
[1] "Risk of James-Stein and Theta: 2.97665537719667"
```

Problem 5

Part (a)

Finding the best linear predictor of Y , which we call $\hat{Y} = \beta_0 + \beta_1 Z_1 + \beta_2 Z_2$ is on that minimizes the mean squared error.

Now, we must find what β_0 , β_1 , and β_2 are. Using Result 7.12 from Johnson and Wichern:

$$\boldsymbol{\beta} = \boldsymbol{\Sigma}_{\mathbf{Z}\mathbf{Z}}^{-1} \boldsymbol{\sigma}_{\mathbf{Z}\mathbf{Y}}$$

$$\beta_0 = \mu_Y - \boldsymbol{\beta}^\top \boldsymbol{\mu}_{\mathbf{Z}}$$

The first thing that we will calculate is the inverse of $\Sigma_{\mathbf{ZZ}}$:

The determinant of $\Sigma_{\mathbf{ZZ}}$ is:

$$\det(\Sigma_{\mathbf{ZZ}}) = 2(1) - 1(1) = 1$$

$$\Sigma_{\mathbf{ZZ}}^{-1} = \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix}.$$

Now, we multiply this quantity with $\sigma_{\mathbf{ZY}}$ to arrive at β

$$\beta = \Sigma_{\mathbf{ZZ}}^{-1} \sigma_{\mathbf{ZY}} = \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} 3 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 - 1 \\ -3 + 2 \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \end{bmatrix}.$$

Therefore, $\beta_1 = 2$ and $\beta_2 = -1$.

The last thing that we need is β_0 :

$$\beta_0 = \mu_Y - \beta^\top \mu_{\mathbf{Z}} = 4 - (2 \cdot 3 + (-1)(-2)) = -4.$$

Finally:

$$\hat{Y} = -4 + 2Z_1 - Z_2$$

Part (b)

The mean squared error of this expression is composed of terms that we have already calculated and terms that we can read off from the covariance matrix:

$$E(Y - (\beta_0 + \beta \mathbf{Z}))^2 = \sigma_{\mathbf{YY}} - \sigma_{\mathbf{ZY}}^\top \Sigma_{\mathbf{ZZ}}^{-1} \sigma_{\mathbf{ZY}}$$

Calculation the mean-squared error, we arrive at:

$$\begin{aligned}
\sigma_{\mathbf{ZY}}^\top \Sigma_{\mathbf{ZZ}}^{-1} \sigma_{\mathbf{ZY}} &= \begin{bmatrix} 3 & 1 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} 3 \\ 1 \end{bmatrix} \\
&= \begin{bmatrix} 3 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ -1 \end{bmatrix} \\
&= (3)(2) + (1)(-1) \\
&= 6 - 1 = 5
\end{aligned}$$

$$\begin{aligned}
E(Y - (\beta_0 + \beta^\top \mathbf{Z}))^2 &= \sigma_{\mathbf{YY}} - \sigma_{\mathbf{ZY}}^\top \Sigma_{\mathbf{ZZ}}^{-1} \sigma_{\mathbf{ZY}} \\
&= 9 - 5 \\
&= \boxed{4}
\end{aligned}$$

Part (c)

To find the population multiple correlation coefficient, we take the positive square root of the ratio between second term of the mean squared error divided by the first:

$$\begin{aligned}
\rho_{Y(\mathbf{Z})}^2 &= \frac{\sigma_{\mathbf{ZY}}^\top \Sigma_{\mathbf{ZZ}}^{-1} \sigma_{\mathbf{ZY}}}{\sigma_{\mathbf{YY}}} \\
&= \boxed{\frac{5}{9}}
\end{aligned}$$

Part (d)

Lastly, the partial correlation coefficient, $\rho_{Y, Z_1 \cdot Z_2}$, is:

$$\begin{aligned}
\rho_{YZ_1 \cdot Z_2} &= \frac{\sigma_{YZ_1} - \frac{\sigma_{YZ_2} \sigma_{Z_1 Z_2}}{\sigma_{Z_2 Z_2}}}{\sqrt{\left(\sigma_{YY} - \frac{\sigma_{YZ_2}^2}{\sigma_{Z_2 Z_2}}\right) \left(\sigma_{Z_1 Z_1} - \frac{\sigma_{Z_1 Z_2}^2}{\sigma_{Z_2 Z_2}}\right)}} \\
&= \frac{3 - \frac{(1)(1)}{1}}{\sqrt{\left(9 - \frac{1^2}{1}\right) \left(2 - \frac{1^2}{1}\right)}} \\
&= \frac{2}{\sqrt{(8)(1)}} \\
&= \frac{2}{2\sqrt{2}} \\
&= \boxed{\frac{\sqrt{2}}{2}}
\end{aligned}$$